Finally, if the government adopts some disastrous policy and then changes its mind, it usually covers itself by blaming the disaster on someone else. Humans learn the hard way to forget facts that might get them in trouble. But how would you train a chatbot to forget that the policy vilified today was actually the official line only a year ago? This is a major technological challenge that dictatorships will find difficult to deal with, especially as chatbots become more powerful and more opaque.

Of course, democracies face analogous problems with chatbots that say unwelcome things or raise dangerous questions. What happens if despite the best efforts of Microsoft or Facebook engineers, their chatbot begins spewing racist slurs? The advantage of democracies is that they have far more leeway in dealing with such rogue algorithms. Because democracies take freedom of speech seriously, they keep far fewer skeletons in their closet, and they have developed a relatively high level of tolerance even to anti-democratic speech. Dissident bots will present a far bigger challenge to totalitarian regimes that have entire cemeteries in their closets and zero tolerance of criticism.

ALGORITHMIC TAKEOVER

In the long term, totalitarian regimes are likely to face an even bigger danger: instead of criticising them, an algorithm might gain control of them. Throughout history, the biggest threat to autocrats usually came from their own subordinates. As noted in chapter 5, no Roman emperor or Soviet premier was toppled by a democratic revolution, but they were always in danger of being overthrown or turned into puppets by their own subordinates. If a twenty-first-century autocrat gives computers too much power, that autocrat might become their puppet. The last thing a dictator wants is to create something more powerful than himself, or a force that he does not know how to control.

To illustrate the point, allow me to use an admittedly outlandish thought experiment, the totalitarian equivalent of Bostrom's paper-clip apocalypse. Imagine that the year is 2050, and the Great Leader is woken up at four in the morning by an urgent call from the Surveillance & Security Algorithm. 'Great Leader, we are facing an emergency. I've crunched trillions of data points, and the pattern is unmistakable: the defence minister is planning to